Advanced Optical Fiber Communications

Problem 10: Dispersion compensation

The silicon dioxide used in an optical fiber has a material dispersion coefficient of 17 ps/(nm km). A step index fiber has a core radius of 4 µm, a numerical aperture of 0.14, and the refractive index of the cladding is 1.44. The used wavelength is 1.55 µm.

a) Calculate the total dispersion of the optical fiber.

b) Now, the total dispersion shall be minimized. Find the core radius, where the waveguide dispersion compensates the material dispersion.

Problem 11: Graded index fiber

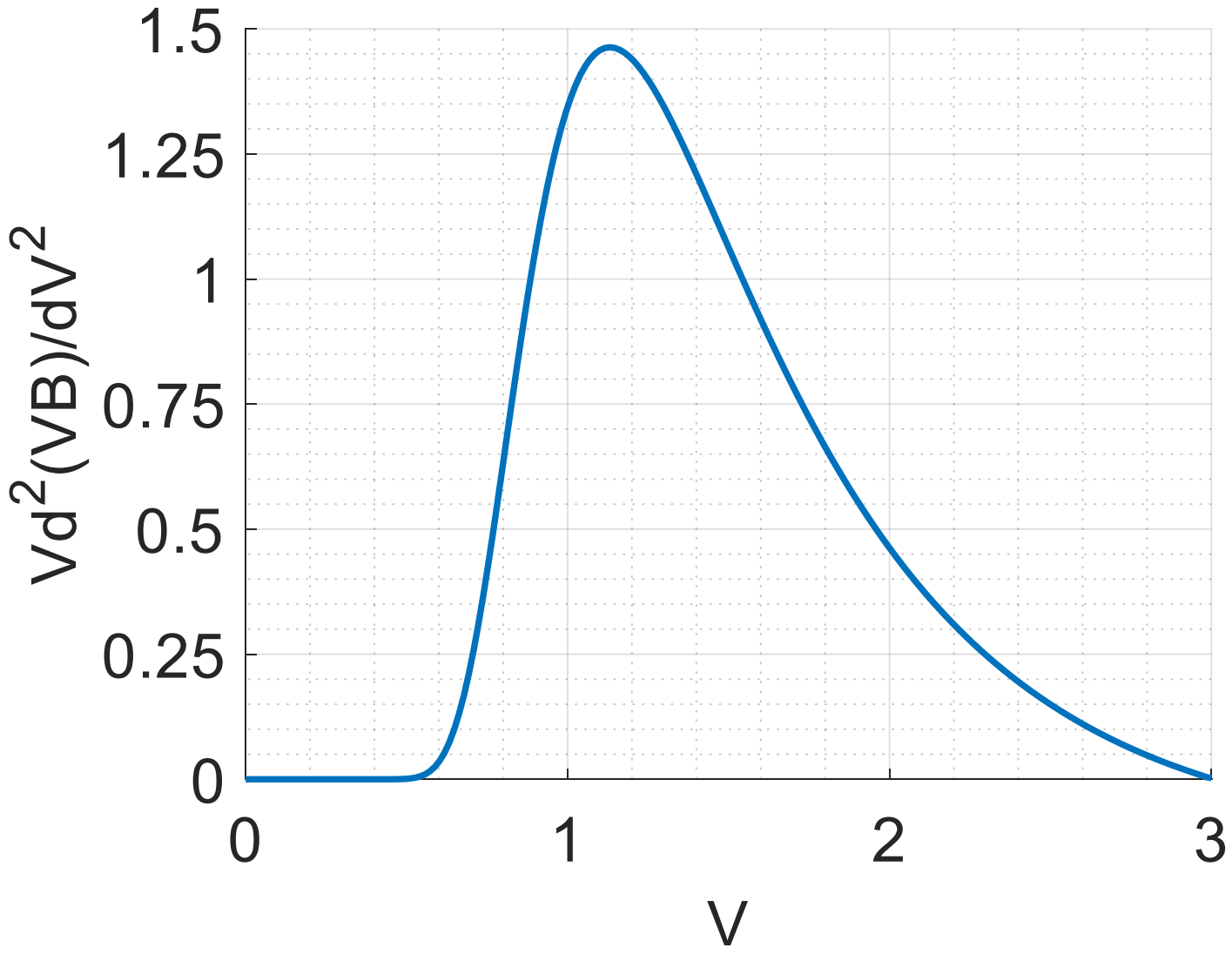
For an optical communication system, a graded index fiber is compared with a step index fiber. The graded index fiber has a parabolic refractive index profile.

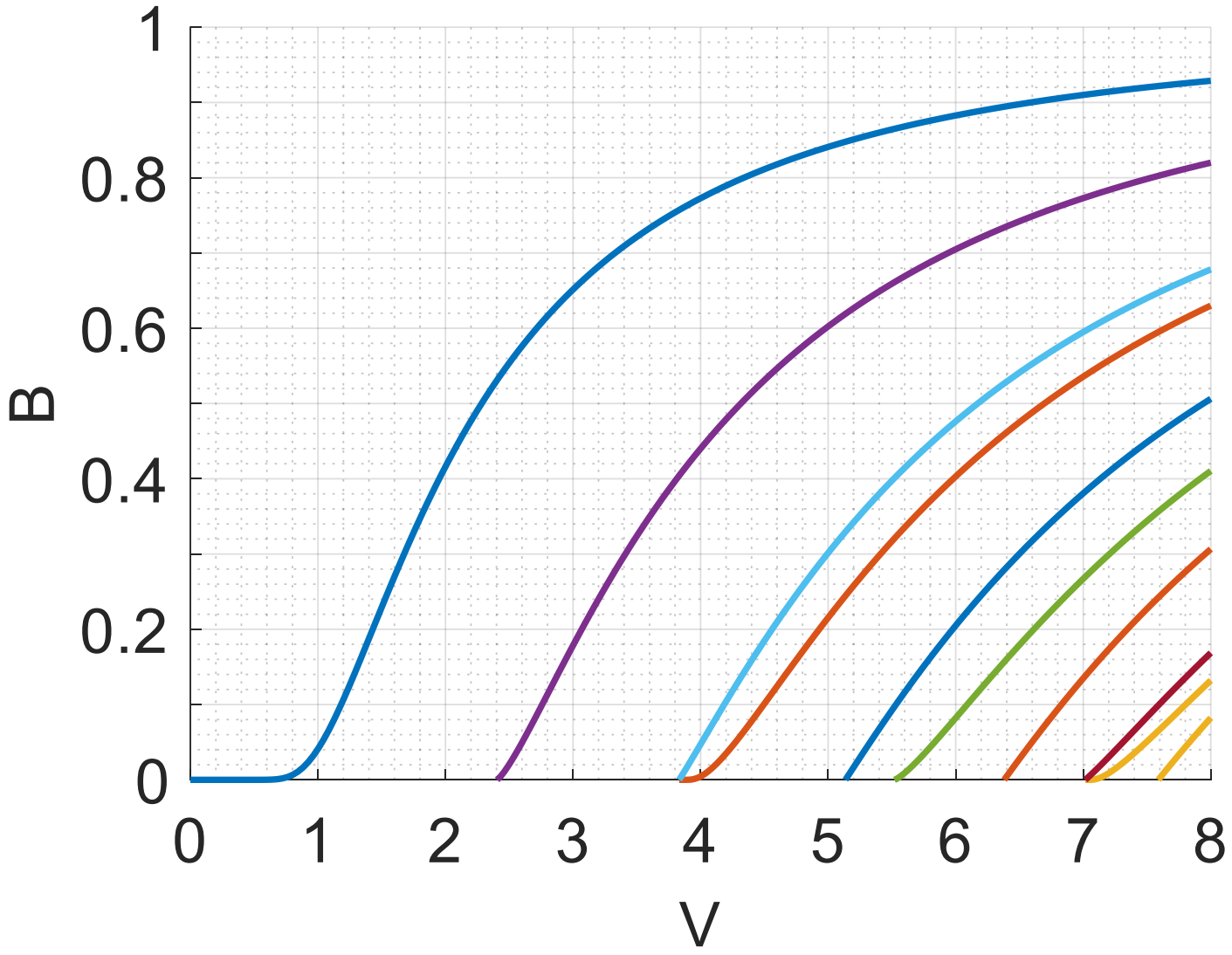
a) Find all supported modes for $V = 8$ for the step index fiber.

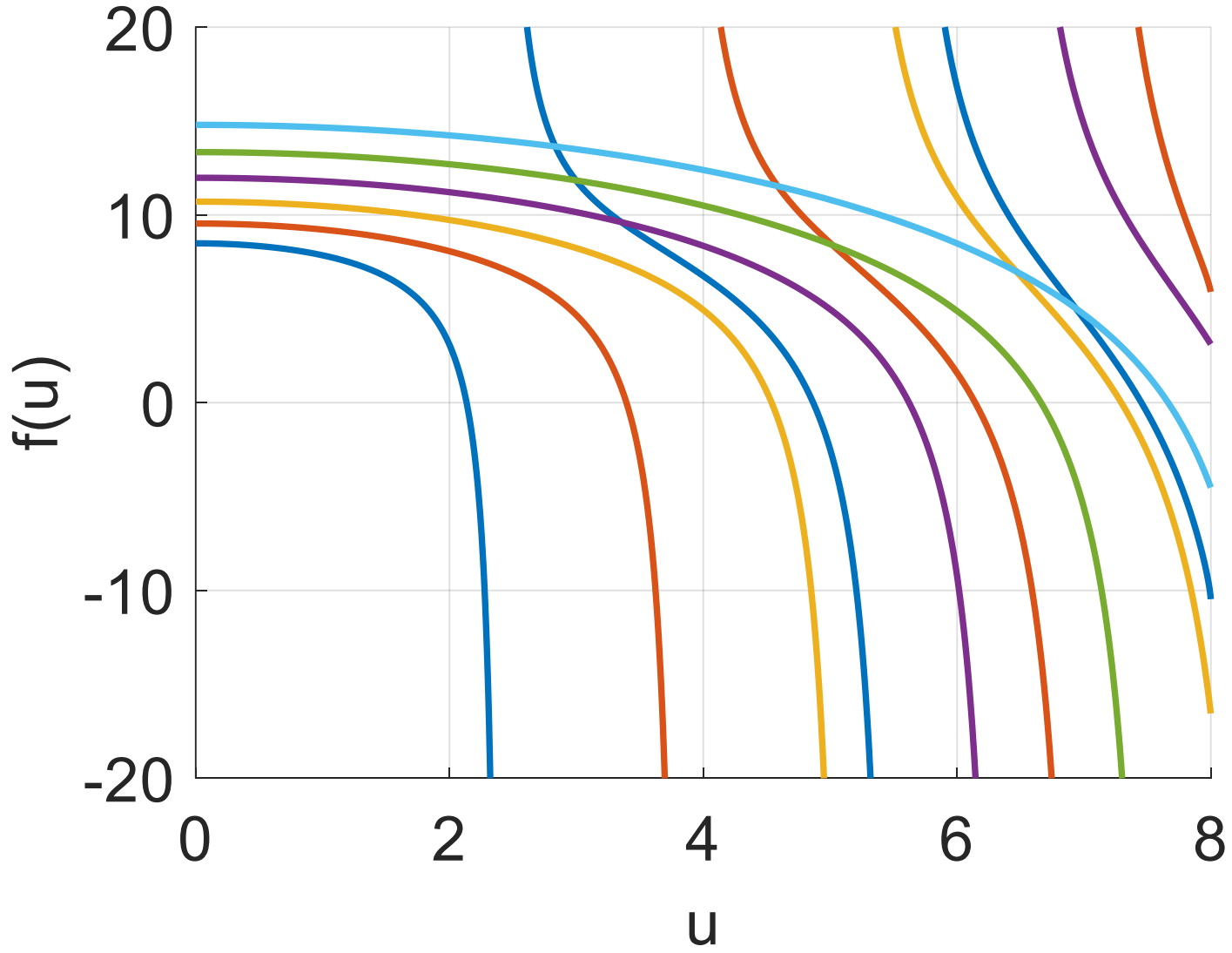
b) Assign the supported modes for the step index fiber to the corresponding mode groups of the graded index fiber. Are all modes of the step index fiber supported by the graded index fiber?

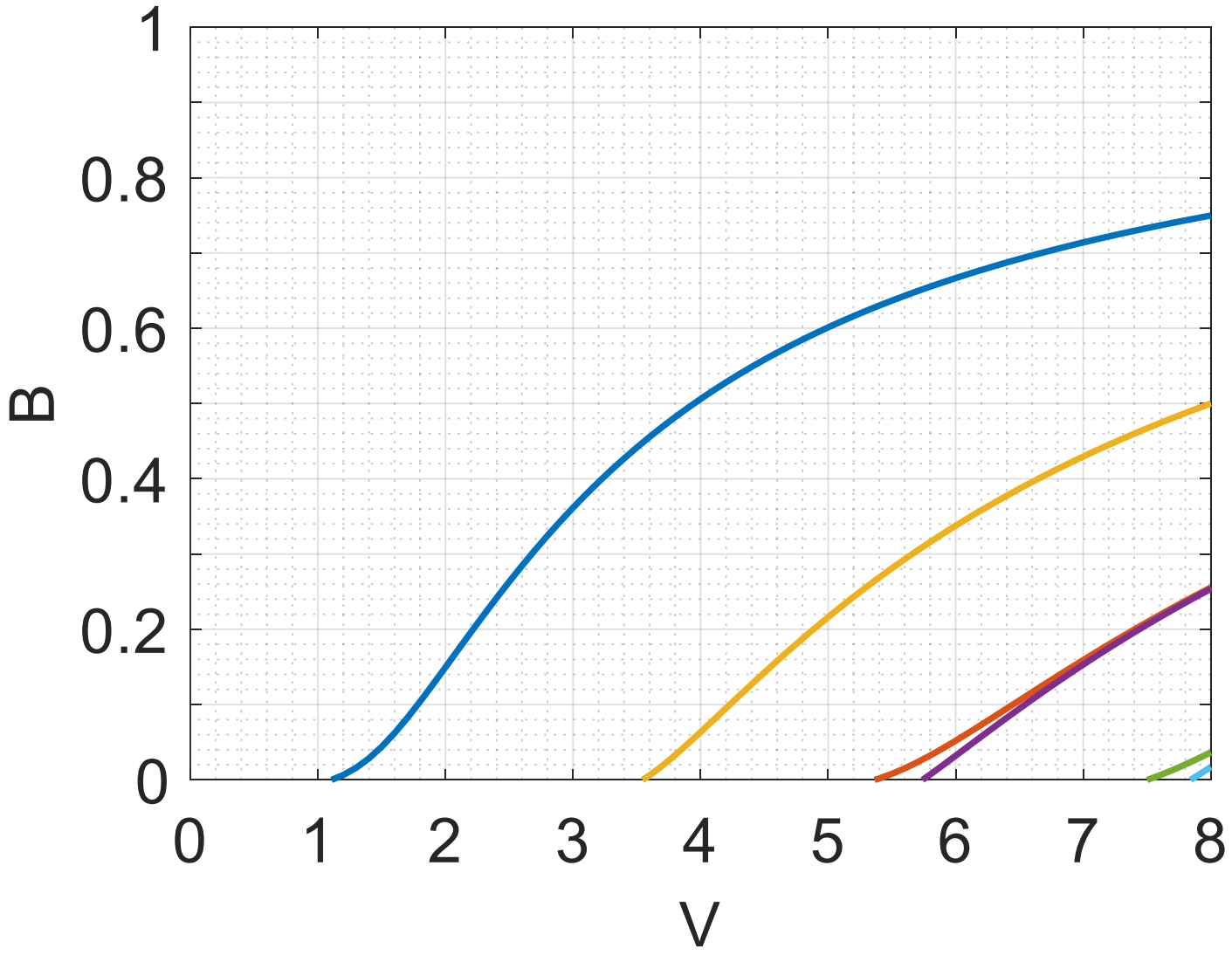
From now on, for both fibers is $V = 11$. The refractive indices are $n_1 = 1.45$ and $n_2 = 1.44$.

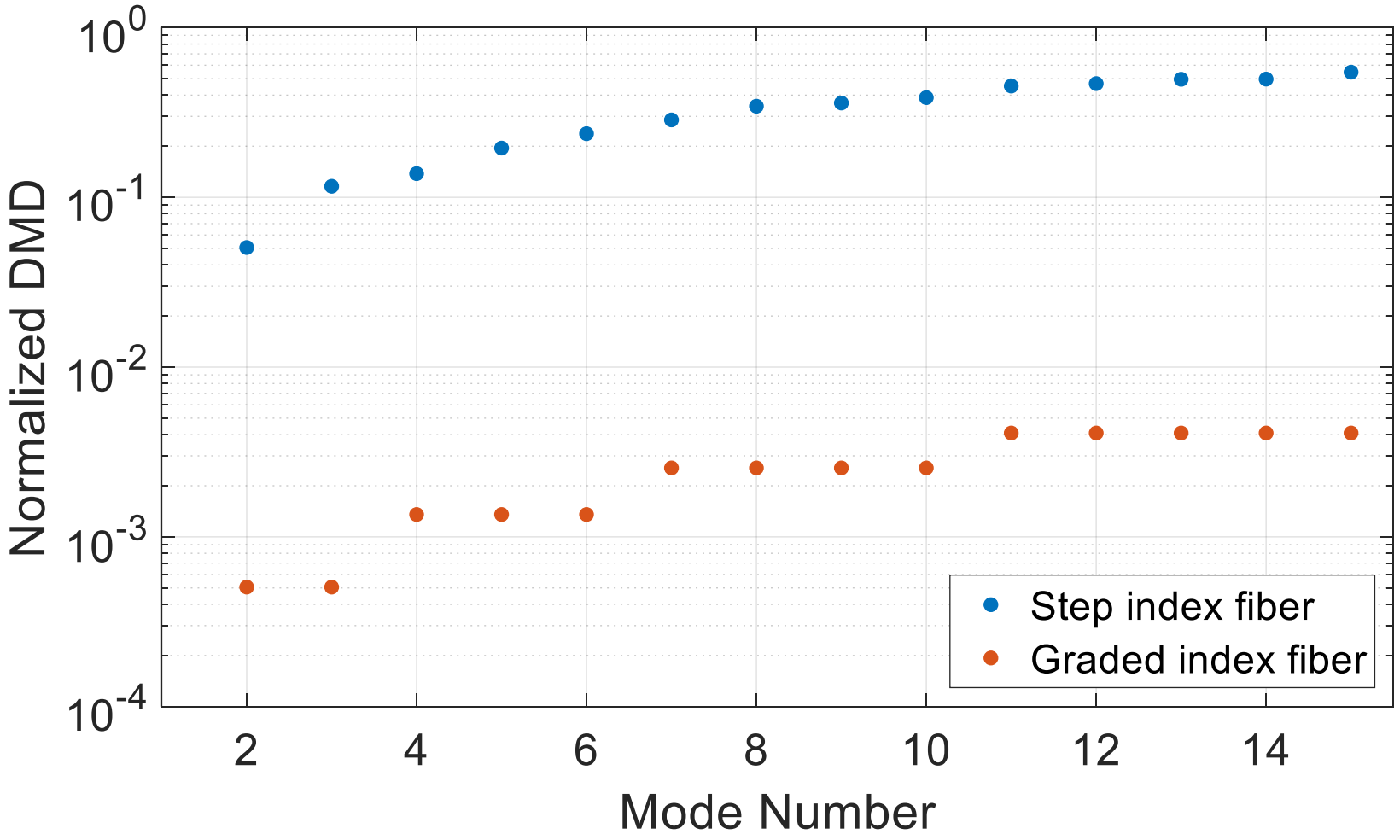
c) The used fiber has a length of 5 km. Calculate the differential mode delay for the 6th mode for the graded index fiber and the step index fiber. What are the maximum possible symbol rates?


1.5
1.25
1
0.75
0.5
0.25
0
$Vd^2(VB)/dV^2$
0
1
2
3
V


B
V
0
0.2
0.4
0.6
0.8
1
0
1
2
3
4
5
6
7
8


20
10
0
-10
-20
f(u)
0
2
4
6
8
u


1
0.8
0.6
0.4
0.2
0
B
0
1
2
3
4
5
6
7
8
V


Normalized DMD
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
2
4
6
8
10
12
14
Mode Number
Step index fiber
Graded index fiber